IAR Workbench

User Guide

SUMMARY: This document provides the installation and configuration required to use IAR Workbench IDE for Atmosic Wireless SoC Series Software Development Kit (SDK) on Windows platform.

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Acronyms and Abbreviations

Acronyms	Definition
EVK	Evaluation Kit
GUI	Graphical User Interface
IDE	Integrated Development Environment
NVDS	Non-Volatile Memory Store
SDK	Software Development Kit

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1. Purpose

This document provides the installation and configuration required to use IAR Embedded workbench for Atmosic SDK on Windows platform.

1.1 Prerequisites

1) Atmosic SDK v5.0.1 or later
2) Atmosic Evaluation Board for ATM3330e/ATM3330 devices, or
3) Atmosic Evaluation Board for ATMx2xx devices and Interface Board v3.x
4) IAR Embedded Workbench v9.20.2.43979

1.2 Toolchain

IAR Embedded Workbench (https://www.iar.com/ewarm) (v9.20.2.43979)

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2. ATM2/ATM3 Platform

2.1 Installation

1) Download the Atmosic SDK and follow the README to setup Universal Serial Bus (USB) device (ignore setup of Universal Serial Bus if using Atmosic SDK Windows Installer)
    a) Libusb-1.0.dll
    b) WinUSB
    c) Viewer for Console Output
    d) Connect mini-USB cable to [USB1> of Interface Board. Confirm `Atmosic RDI USB1` is showing in Windows device manager under USB devices
    e) Connect mini-USB cable to [USB0> of Interface Board. Confirm `Atmosic RDI USB0` is showing in Windows device manager under USB devices

2) I-jet Probe/debugger
    a) Insert the I-jet device.
    b) After successful installation, `IAR Systems USB Probes` should display under Device Manager as shown in Figure 1.

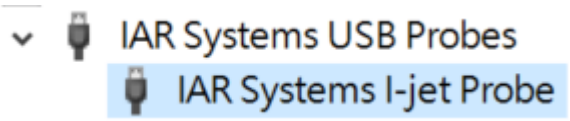


Figure 1 - I-jet Installation Successful

    c) Pin Connection:
        i) Atmosic Interface Board V3.x, see Figure 2:
        Remove JP1, JP2 and J3 jumpers on the interface board.

Use the attached `MIPI-10 JTAG cable` to connect to the J-Link socket of the interface board.



Figure 2 - Interface Board V3.x

3) Install GNU Tools ARM Embedded (can ignore this step if using Atmosic SDK Windows Installer) and add the toolchain path to the user or system path

4) Add a system environment variable: (Replace <SDK installation folder> name with actual installation folder name of Atmosic SDK)
   a) System variable: ATM_SDK_MSYS2
   b) Value: <SDK installation folder>\tools\msys2-base-x86_64-20220128

5) Edit system environment Path to add gcc toolchain in path. (Replace <SDK installation folder> with actual folder name)
   a) Add <SDK installation folder>\tools\gcc-arm-none-eabi-10.3-2021.07\bin

6) Reboot for system environment

7) Flash Loader
   Copy atmx_flash.board, atmx_flash.flash, atmx_flash.mac, atmx_flash.out from <SDK installation folder>\tools\iar to <IAR installation folder>\arm\config\flashloader\Atmosic

See Generate IAR workspace section to generate IAR workspace.

2.2 Generate IAR workspace

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Please refer to IDE Auxiliary Flash Programming tool User Guide, How to Generate IDE Project section to generate IAR workspace.

After generating the workspace successfully, it can be found in the following folder:

<Atmosic SDK>\platform\atm2\ATM2xxx-xxx\example\xxx\iar

<Atmosic SDK>\platform\atm3\ATM2xxx-xxx\example\xxx\iar

Open workspace file in iar folder, ie: example\xxx\iar\<xxx>.eww

2.3 Build and Program

1) Program Flash
    a) With Non-Volatile Data Store (NVDS) data for each example (refer to Program Flash with NVDS section)

    b) Without NVDS for each example

2) Edit/Compile/Program/Debug Code at IAR IDE

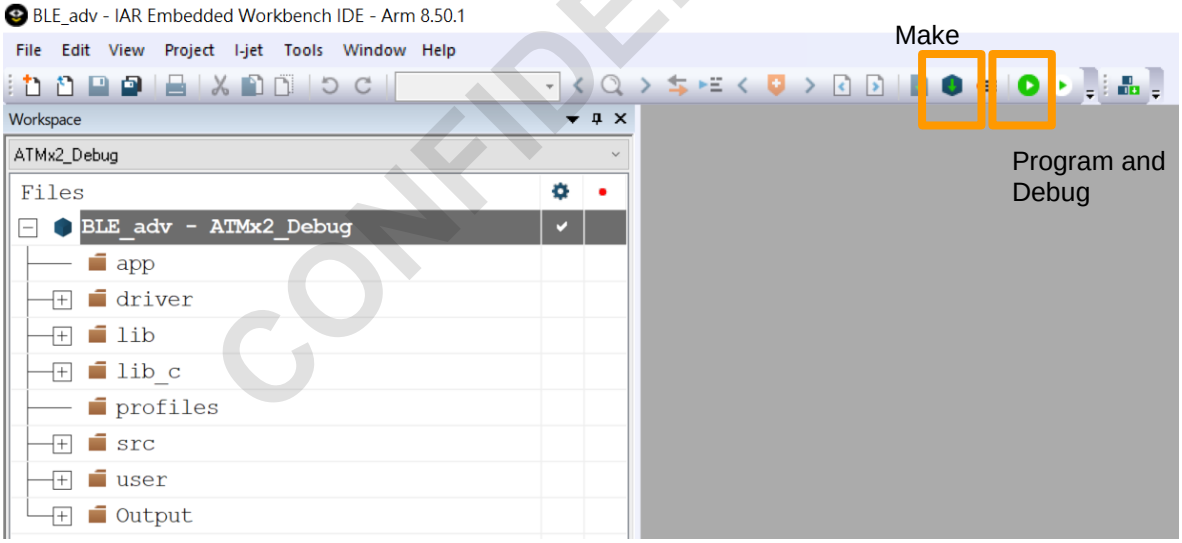


Figure 3 - IAR Workbench GUI

3) Run time debugger, Figure 4 shows the layout of the debugger session as an example

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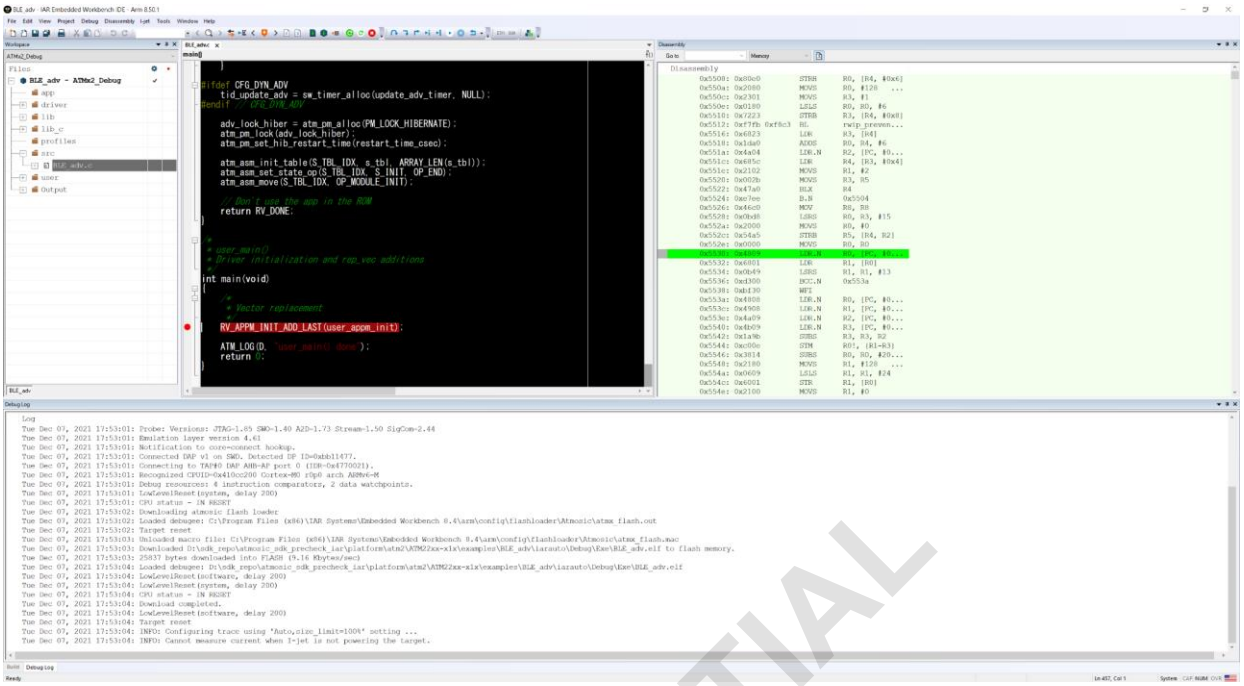


Figure 4 - Debug Session

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2.4 Set Flash Loader

When creating a new IAR project, it needs to override the default .board file.
Path is $TOOLKIT_DIR$\config\flashloader\Atmosic\atmx_flash.board. See Figure 5.

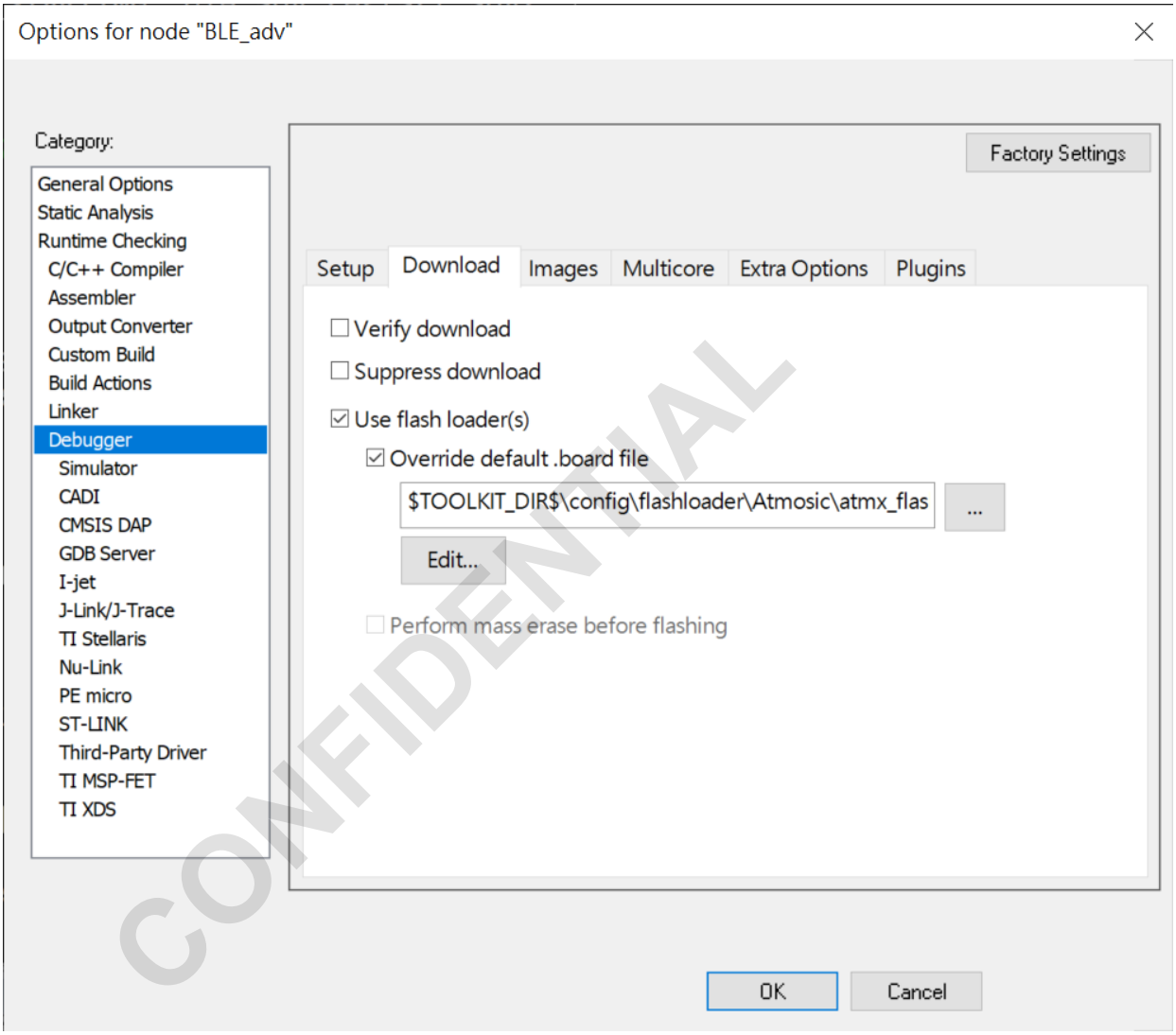


Figure 5 - Set Flash Loader

2.5 Program Flash with NVDS

Each iar workspace folder will include the flash_nvds.bin file.

To program NVDS, set flash_nvds.bin as an input and program the whole image. See Figure 6.

1) Go to linker setting page: Project Options →Linker →Input
    a) Keep symbols: NVDS
    b) Raw binary file: $PROJ_DIR$\flash_nvds.bin

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c) Symbol:NVDS, Section:.NVDS, Align:8

2) Rebuild whole project

3) Program image

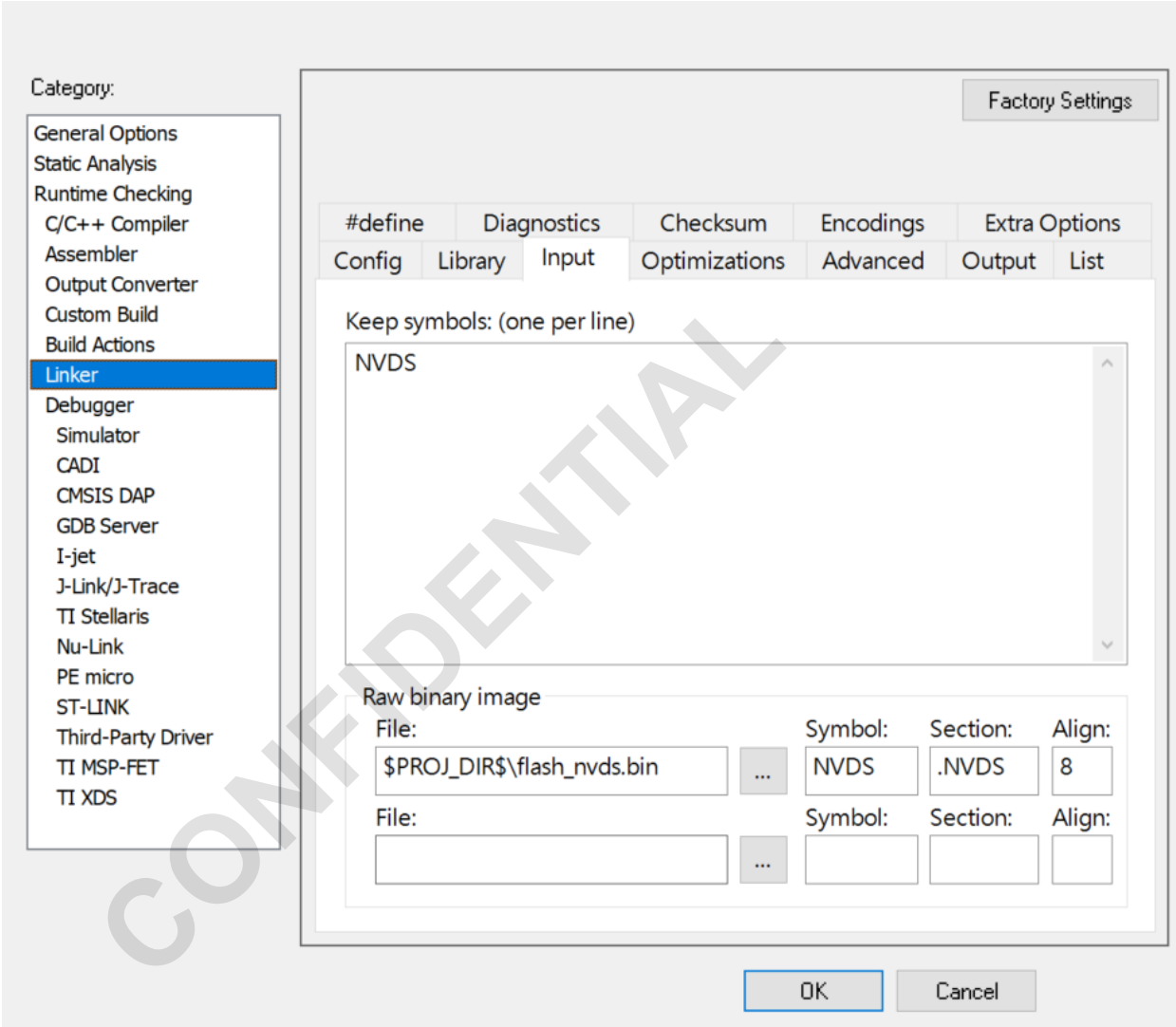


Figure 6 - Add flash_nvds.bin

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3. ATM33/ATM33e Platform

3.1 Installation

1) Download the latest Atmosic SDK and follow the README to setup USB device.

   Connect USB cable to the USB Connector on the ATM3330e/ATM3330 Evaluation Board, see Figure 7. Confirm JLink CDC UART Port and J-Link driver are showing in Windows device manager.



Figure 7 - ATM33/e Evaluation Board

2) Install GNU Tools ARM Embedded (can ignore this step if using Atmosic SDK Windows Installer) and add the toolchain path to the user or system path.

3) Add a system environment variable: (Replace <SDK installation folder> name with actual installation folder name of Atmosic SDK)
   System variable: ATM_SDK_MSYS2
   Value: <SDK installation folder>\tools\msys2-base-x86_64-20190524

4) Edit system environment “Path” to add gcc toolchain in path. (Replace <SDK installation folder> with actual folder name)
   Add <SDK installation folder>\tools\gcc-arm-none-eabi-10.3-2021.07\bin

5) Reboot for system environment

6) Flash Loader
   Copy all files from <SDK installation folder>\tools\iar to <IAR installation folder\arm\config\flashloader\Atmosic>

7) Refer to Generate IAR workspace to generate IAR workspace.

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8) Two options to build and program firmware

Option 1: Generate IAR workspace without USE_MCUBOOT

- Refer to Build Example and Program (without US_MCUBOOT) to build and program bootloader
- Refer to Program Image without NVDS and Program Image with NVDS to build and program application

Option 2: Generate IAR workspace with USE_MCUBOOT

- Edit and build the project to generate application image
  - ***Note**: if the bootloader was updated, then it needs to follow section 3.3 to link right Secure_Functions.o.*
- Refer to IDE Auxiliary Flash Programming tool User Guide to program generated image.

3.2 Generate IAR workspace

Please refer to section 8 of **IDE Auxiliary Flash Programming Tool User Guide** to generate IAR workspace.

After generating successfully, the workspace can be found in following folder:

<Atmosic SDK>\platform\atm33\ATM33xx-5\example\xxx\iar

Open workspace file in iar folder, ie: example\xxx\iar\<xxx>.eww

3.3 Build and Program Bootloader

Bootloader is an example to boot the system and it will generate an object file called Secure_Functions.o for applications to link and to use the secure entry functions. So it's important to know what Secure_Functions.o was programmed into the device. Application must link with proper Secure_Functions.o.

By default, the project will automatically use a script called prebuild.bat(located in the workspace folder) to generate a bootloader image in SDK\platform\atm33\ATM33xx-5\example\bootloader\ and link it with the application.
If the bootloader is needed to build using IAR, please refer to following steps to build and link the Secure_Functions.o.

IAR workspace to generate bootloader

1) Run Setup IDE Environment and ensure J-Link driver is showing in Windows device manager under USB devices.

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2) Generate IAR workspace, (please refer to Generate IAR workspace) and Open IAR workspace at SDK\platform\atm33\ATM33xx-5\example\bootloader\iar\bootloader.eww

3) Build and Program bootloader.

4) Modify the link option in the application workspace.
    a) Project -> Options -> Linker -> Extra Options
    b) Replace $PROJ_DIR$\..\..\bootloader\Secure_Functions.o -> $PROJ_DIR$\..\..\bootloader\iar\Debug\Exe\Secure_Functions.o

3.4 Build Example and Program (without USE_MCUBOOT)

1) Run Setup IDE Environment and ensure J-Link driver is showing in the Windows device manager under USB devices.

2) Edit/Compile/Program Code in IAR IDE, see Figure 8.

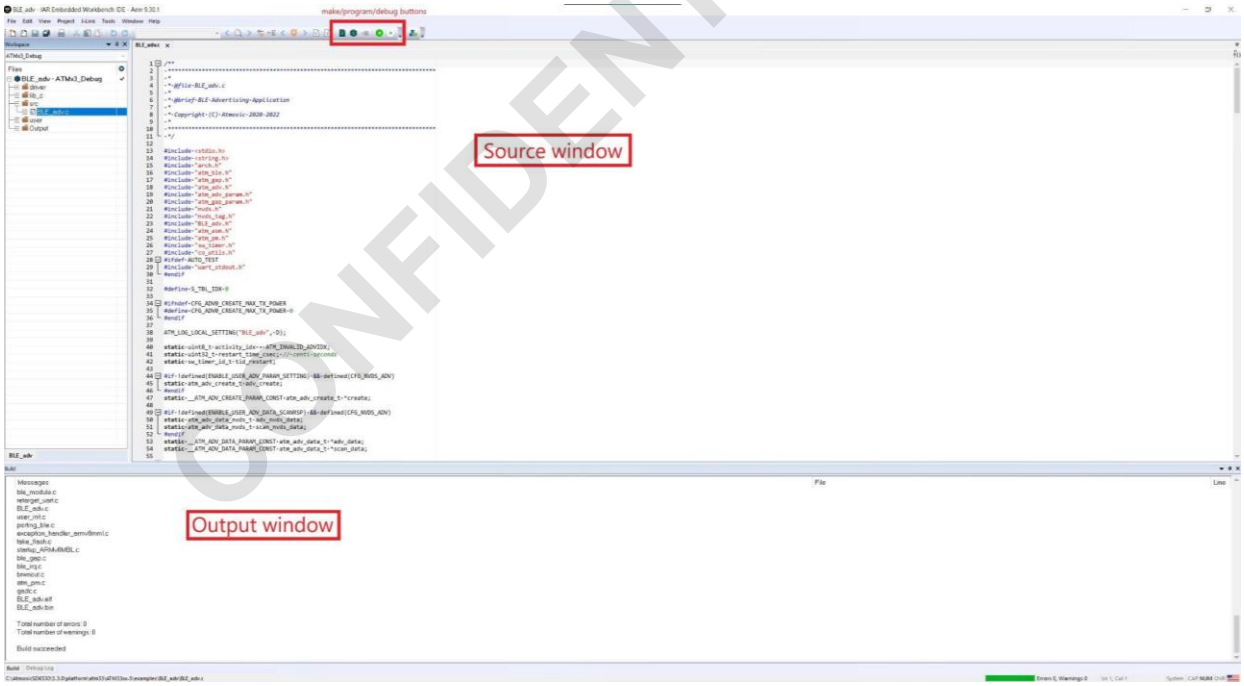


Figure 8 - IAR Workbench GUI

3) Program and Debug image, please refer to Figure 12 for programming image with NVDS
    a) Press the Download and Debug button on the toolbar or (Project → Download → Download and Debug), see Figure 9.
    b) Runtime debug, see Figure 10.

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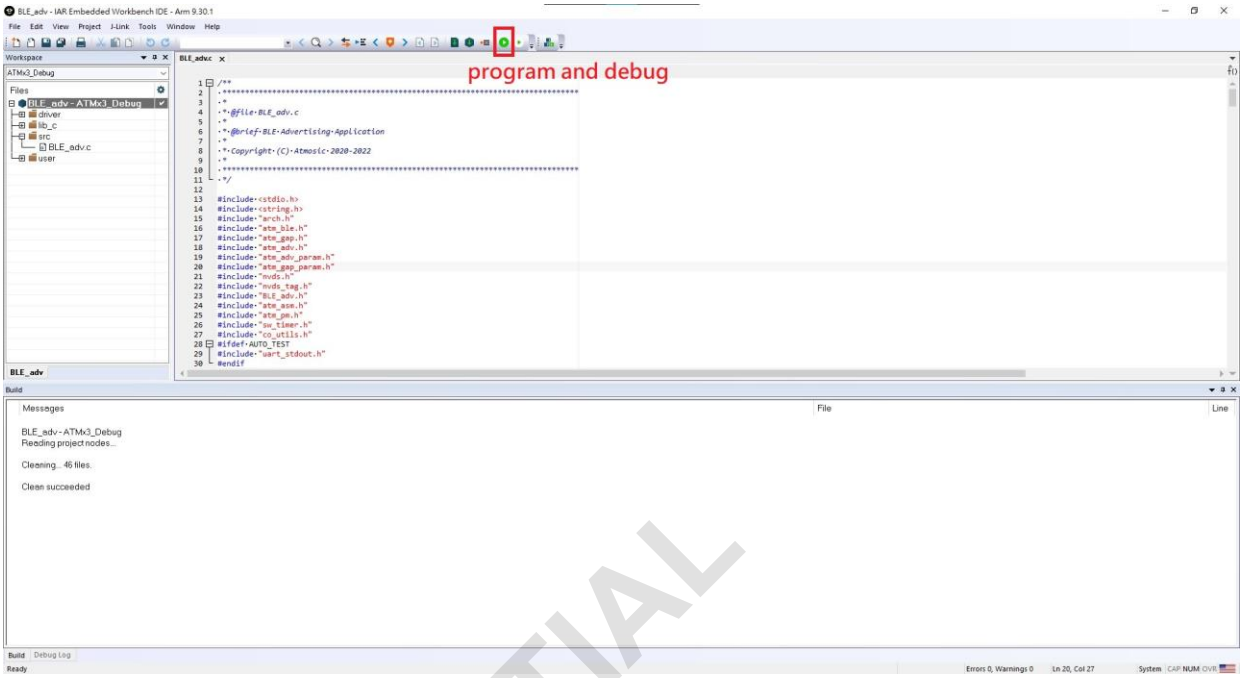


Figure 9 - Download and Debug

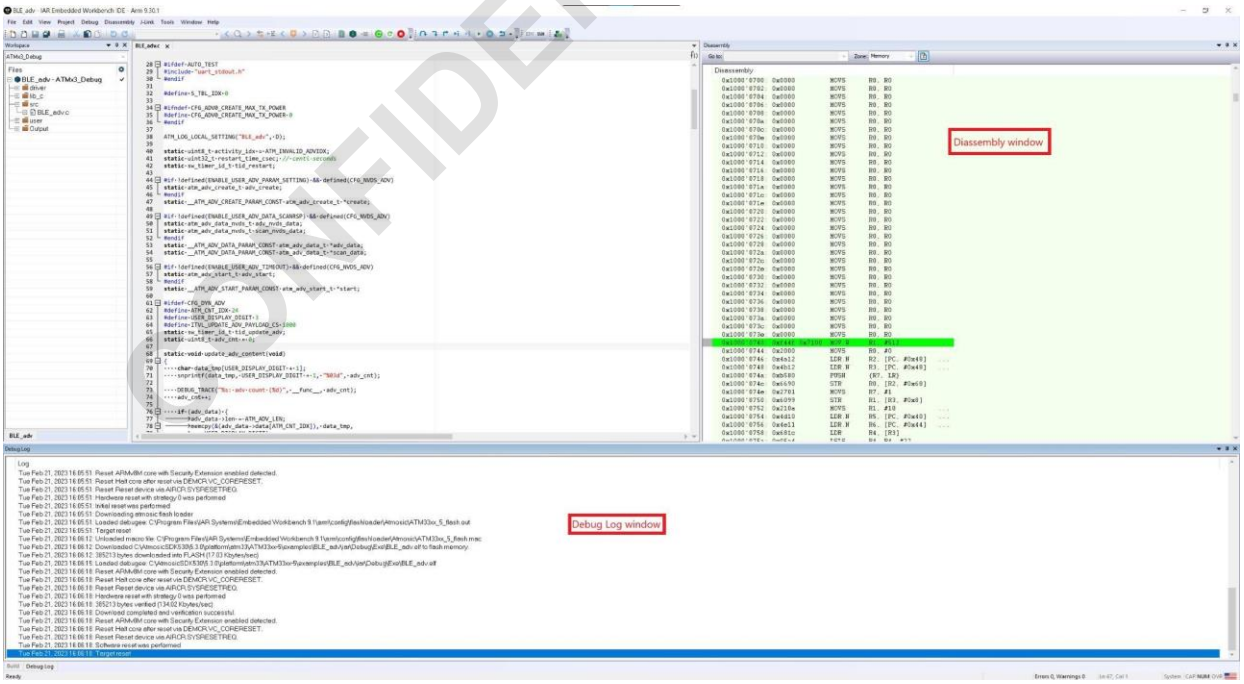


Figure 10 - Runtime Debug

4) See the log on Debug port, see Figure 11.

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```
@000000b4 PSEQ STATUS=0x1
@000000ff RESET_SYNDROME=0x1
@00000156 PMU WKUP_DET = 0
@000001a5 boot_status = 1000000
@00000203 Cold boot
@0000023f  Power on Reset
@000002ba pmu_init
@000002ec bp_freq is 16000000
@00000340 pmu_nonharv
@00000381 SDK Version: 5.3.0
@000003cd APP Version: 0.0.0.9
WARNING: OTA update not supported with this image
NVDS_RRAM: 4e 56 44 53 06 06 27 00 00 00 00 01 02 00 00 00 ...
@0000054d [   BLE_adv][D]: user_main() done
@0000060a rw ISRs configured
@0000066d [   BLE_adv][D]: ble_adv_init: NVDS tag for adv timeout param not
found. Using default
@00000781 rwip_init() done
@000007ca Entering main loop
@00000899 [   atm_adv][D]: Adv dur 0(unit:10ms) max_adv_evt 0 (timeout 0ms)
@00000992 [   BLE_adv][D]: adv_state = 2
@00000a33 [   BLE_adv][D]: adv_state = 4
@00000ac6 [   BLE_adv][D]: adv_state = 6
@00000b68 [   atm_adv][D]: Adv0: ON
@00000be1 [   BLE_adv][D]: adv_state = 9
```

Figure 11 - BLE_adv Example Log

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3.5 Program Image with NVDS (without USE_MCUBOOT)

To program with NVDS, it needs to add the settings below. See Figure 12.

[Steps for add settings to program NVDS>

1) Go to linker setting page: Project Options →Linker →Input
    a) Keep symbols: NVDS
    b) Raw binary file: $PROJ_DIR$\flash_nvds.bin
    c) Symbol:NVDS, Section:.NVDS, Align:8

2) Rebuild the whole project and the flash_nvds.bin will be linked into the final ELF file.

3) Program image

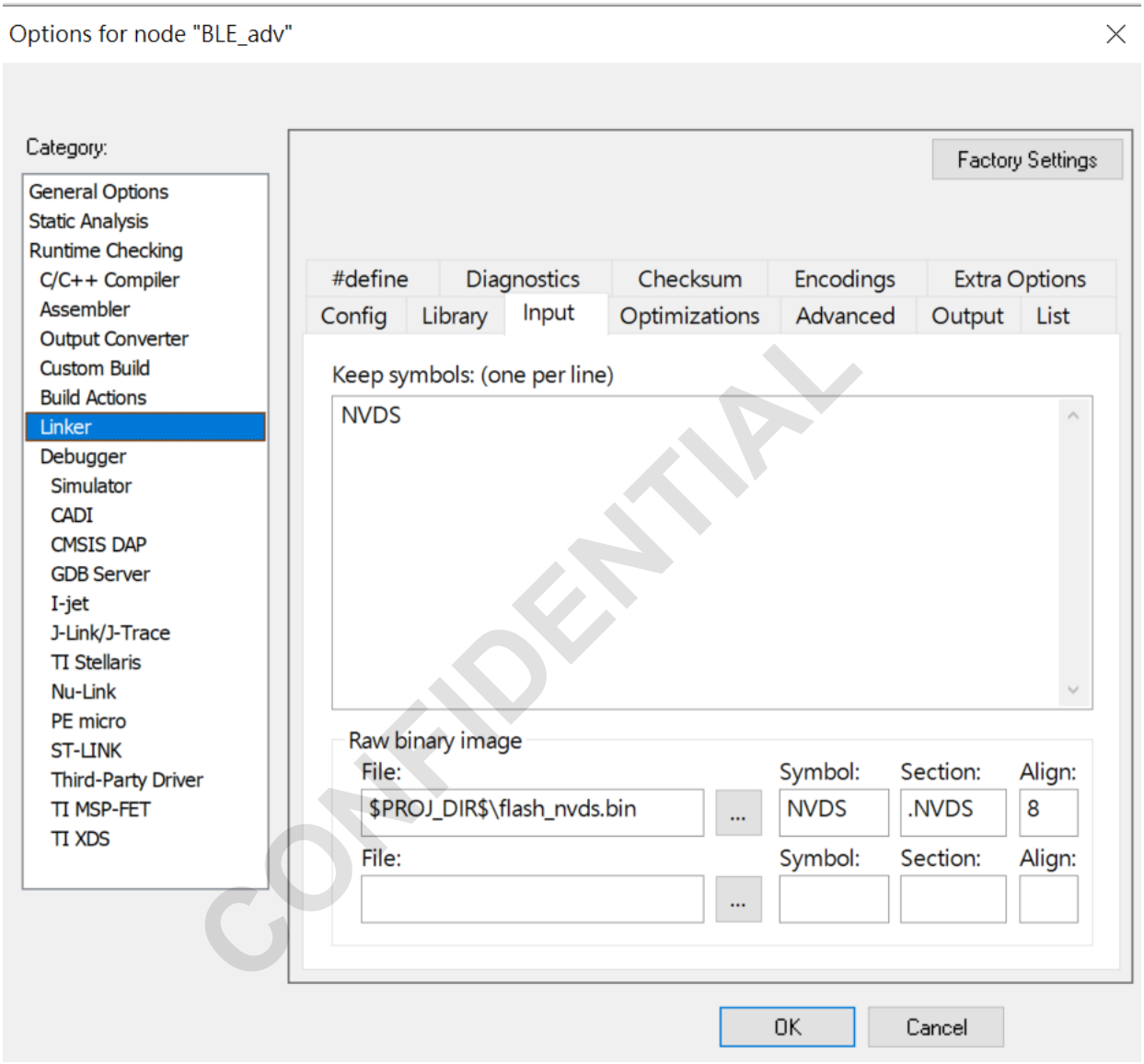


Figure 12 - Program without NVDS (without USE_MCUBOOT)

Reference Documents

Title	Document Number
IDE Auxiliary Flash Programming Tool User Guide	ATM-UGIDEFPT

Revision History

Date	Version	Description
March 17, 2023	0.61	Updated Generate IAR workspace, Program Flash with NVDS, Installation, Build and Program Bootloader, Figure 7 - ATM33/e Evaluation Board, Program Image with NVDS
May 13, 2022	0.60	Initial version created for SDK 5.1 which supports ATM2/ATM3 and ATM33 platforms.

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